

Residual diagonals for $\lambda = (\hat{\lambda}, 1^q)$:
 refutation of the Kazhdan–Lusztig cell hypothesis
 and an empirical column-1 SC discriminator
 at $\hat{\lambda} = (3, 3)$

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Abstract

The SC-at-1 trivial reduction closes 378/514 of the empirically tested converse Diagnostic 4 cases. The remaining open core is the *SR-at-1 residual*: zero diagonals $p_T(q) = 0$ for which the only column-strict pairs in T occur at indices $i \geq 2$. We make three contributions. First, we test and refute the Dai–Zhang-inspired hypothesis that the SR-at-1 residual is a union of cells in the Hoefsmit/Murphy W -graph or the descent partition; on $\lambda = (3, 3, 1, 1, 1)$, two descent classes each contain one residual and one nonresidual member. Second, we propose and verify (on 50/50 SR-at-1 tableaux of $(3, 3, 1, 1, 1)$ with 0 errors) an explicit combinatorial discriminator built from *column-1* SC pairs at the arithmetic-progression indices $\{3, 5, 7\}$. Third, we identify a sign-cancellation phenomenon: three nonresidual tableaux with “double SC at γ -position” (entries 7, 8, 9 all stacked in column 1) escape the discriminator’s γ -clause, and the non-continuation clause is essential. We retract the earlier Ω^* -only count of 12 for this shape: all 35 residual zero diagonals at $(3, 3, 1, 1, 1)$ are killed by Ω and Ω^* simultaneously.

1 Background and the residual problem

1.1 Setting and notation

Fix $\lambda = (\hat{\lambda}, 1^q)$ with $\ell(\hat{\lambda}) \geq 2$, every row $\hat{\lambda}_r \geq 2$, and $\tau(\hat{\lambda}) := \max(0, q + 2\ell(\hat{\lambda}) - 1 - |\hat{\lambda}|) = 0$. Let $n = |\lambda|$, $\ell = \ell(\lambda)$. Let V^λ be the irreducible $\mathcal{H}_q(\mathfrak{S}_n)$ -module in the Hoefsmit (seminormal) basis $\{v_T : T \in \text{SYT}(\lambda)\}$; the Hecke generators T_i satisfy $(T_i - q)(T_i + 1) = 0$ and we set $S_i := T_i + 1$. Define the chain factors and the diagnostic operator

$$\begin{aligned} R'_k &:= S_{k-1}S_{k-2} \cdots S_2S_1, \\ L_k &:= S_1S_2 \cdots S_{k-2}S_{k-1}, \\ \Omega &:= R'_{\ell+1}R'_{\ell+2} \cdots R'_n, \\ \Omega^* &:= L_nL_{n-1} \cdots L_{\ell+1}. \end{aligned}$$

The diagonal probe is $p_T(q) := \langle v_T, \Omega v_T \rangle \in \mathbb{Q}(q)$ with respect to the Hoefsmit pairing.

Definition 1. For $T \in \text{SYT}(\lambda)$, an *SC pair* at index i is a pair $(i, i + 1)$ with the cell containing $i + 1$ directly below the cell containing i (same column). An *SR pair* is the row analogue. We say T is *SC-at-1* if the cell containing 2 is $(2, 1)$, otherwise (since 2 lies at $(1, 2)$ or $(2, 1)$ in any SYT) T is *SR-at-1*.

1.2 The two facts in hand

Two structural results frame the residual question.

- **Diagnostic 4 dichotomy** (forward direction proved 2026-05-18). $p_T(q) = 0 \Leftrightarrow \Omega v_T = 0$ or $\Omega^* v_T = 0$. The forward direction is proved by the two-sided sign-kill identity; the converse is open.
- **Universal SC necessity** (proved 2026-05-18). If $p_T(q) = 0$ then T has at least one SC pair (Hoefsmit positivity rigidity on the diagonal path). The converse fails: T may have an SC pair with $p_T \neq 0$.

1.3 The SC-at-1 trivial reduction

Yesterday's note `2026-05-19-sc-at-1-trivial-reduction.tex` showed that if T is SC-at-1 then $S_1 v_T = 0$, hence $R'_n v_T = 0$ (since S_1 is the rightmost factor of R'_n), hence $\Omega v_T = 0$. This closes $378/514 \approx 73.5\%$ of the converse cases in our 12-shape corpus.

1.4 The residual: SR-at-1

The remaining open problem isolates the 136 zero diagonals T with T SR-at-1, i.e. entries 1, 2 both in row 1. The algebraic obstacle is that for SR-at-1 tableaux, S_1 does not annihilate v_T , so the one-line argument fails. The full converse of Diagnostic 4 reduces to:

For every $T \in \text{SYT}(\lambda)$ with T SR-at-1 and $p_T(q) = 0$, either $\Omega v_T = 0$ or $\Omega^* v_T = 0$.

This is the SR-at-1 residual problem. At $\lambda = (3, 3, 1, 1, 1)$, the residual contains 35 tableaux among 50 SR-at-1 tableaux (the other 15 are SR-nonzero).

2 The cell hypothesis

2.1 Motivation

Dai–Zhang [1] construct *recursive Gelfand W -graphs* for the symmetric group, refining classical Kazhdan–Lusztig cells. In the Hoefsmit/Murphy basis of V^λ , several natural partitions on $\text{SYT}(\lambda)$ arise:

- the Murphy graph (SCC partition on the basis transition graph),
- the *descent partition*: two SYTs are equivalent iff they have the same descent set $D(T) := \{i : i + 1 \text{ is strictly below } i \text{ in } T\}$,
- the dual-equivalence (Assaf) partition,
- the (recursive) Gelfand W -graph cells of [1].

The hopeful version of the SR-residual hypothesis runs:

Conjecture 2 (Cell hypothesis). *There exists a partition Π of $\text{SYT}(\lambda)$ arising from a Kazhdan–Lusztig/Murphy-type W -graph such that the SR-at-1 residual is a union of blocks of Π .*

If true, Conjecture 2 would yield a uniform proof of the SR-at-1 converse by cell-induction [1]. The *descent partition* is the finest natural partition compatible with KL left cells, and the Murphy/Knuth dual-equivalence partition within a fixed shape is one giant class (Haiman). So the descent partition is the strongest test of Conjecture 2.

3 Refutation at $\lambda = (3, 3, 1, 1, 1)$

3.1 Setup

We take $\lambda = (3, 3, 1, 1, 1)$, so $\hat{\lambda} = (3, 3)$, $q = 3$, $\ell(\hat{\lambda}) = 2$, $|\hat{\lambda}| = 6$, $\tau = 0$, $n = 9$, and

$$\Omega = R'_3 R'_4 R'_5 R'_6 R'_7 R'_8 R'_9.$$

The Specht module V^λ has $N = 120$ basis vectors. Of these, 70 are SC-at-1 (each closed by the trivial reduction) and 50 are SR-at-1. Among the SR-at-1 tableaux, 35 are residual ($p_T = 0$) and 15 are nonresidual ($p_T \neq 0$). All counts and diagonal probes use the symbolic Hoefsmit basis specialised at $q \in \{2, 3, 5, 7\}$; agreement at four distinct positive rationals fixes the diagonal entry to 0 in $\mathbb{Q}(q)$ at the degrees encountered here.

3.2 The decisive descent blocks

The descent partition on $\text{SYT}(3, 3, 1, 1, 1)$ has 95 blocks (21 non-singleton). Two non-singleton descent blocks each contain one SR-residual and one SR-nonresidual member:

Block $D = (2, 4, 5, 7, 8)$ **with members** $\{T_{11}, T_{27}\}$:

- $T_{11} = (1, 2, 4 \mid 3, 5, 7 \mid 6 \mid 8 \mid 9)$. SC list = $[8]$. SR list = $[1]$. $p_{T_{11}} \neq 0$.
- $T_{27} = (1, 2, 7 \mid 3, 4, 8 \mid 5 \mid 6 \mid 9)$. SC list = $[5, 7]$. SR list = $[1, 3]$. $p_{T_{27}} = 0$.

Block $D = (2, 4, 5, 6, 8)$ **with members** $\{T_{12}, T_{29}\}$:

- $T_{12} = (1, 2, 4 \mid 3, 5, 8 \mid 6 \mid 7 \mid 9)$. SC list = $[\]$. SR list = $[1]$. $p_{T_{12}} \neq 0$.
- $T_{29} = (1, 2, 8 \mid 3, 4, 9 \mid 5 \mid 6 \mid 7)$. SC list = $[5, 6, 8]$. SR list = $[1, 3]$. $p_{T_{29}} = 0$.

In each block, both members are SR-at-1 (entries 1, 2 in row 1) and share the same descent set, yet exactly one is residual. The nonresidual member (T_{11} or T_{12}) has no SC pair below index 8; the residual member has SC pairs at indices ≤ 6 .

3.3 Auxiliary checks

The Murphy SCC partition on $\text{SYT}(3, 3, 1, 1, 1)$ in the Hoefsmit basis is the trivial giant component (one block). Thus the Murphy partition contains the residual set vacuously but provides no separating power. Among partitions strictly coarser than the trivial one, the descent partition is the finest reasonable cell candidate, and it already fails.

Theorem 3 (Refutation of the cell hypothesis at $(3, 3, 1, 1, 1)$). *The SR-at-1 residual on $\lambda = (3, 3, 1, 1, 1)$ is not a union of descent classes. In particular, it is not a union of blocks of any partition coarser than or equal to the descent partition; this rules out the Murphy and dual-equivalence partitions on this shape as candidates for Conjecture 2.*

Proof. The blocks $\{T_{11}, T_{27}\}$ and $\{T_{12}, T_{29}\}$ each contain exactly one residual member. A partition cannot have the residual set as a union of blocks if some block splits between residual and nonresidual. $\square$

Remark 4. Theorem 3 does not rule out finer KL-recursive partitions (e.g. Gelfand W -graph cells from [1] or asymptotic two-sided cells). It does say that the naive partitions accessible from the Hoefsmit basis alone do not encode the residual stratum, and any uniform cell-theoretic proof must descend below the descent partition.

4 The column-1 SC discriminator at $\hat{\lambda} = (3, 3)$

4.1 Statement

For each $T \in \text{SYT}(3, 3, 1, 1, 1)$, write $T(r, c)$ for the entry in row r , column c (so column 1 has entries $T(1, 1) = 1, T(2, 1), T(3, 1), T(4, 1), T(5, 1) = 9$, with the last forced because 9 is the largest entry and $(5, 1)$ is the only extremal cell with row index 5).

Theorem 5 (Empirical, $\lambda = (3, 3, 1, 1, 1)$). *For every $T \in \text{SYT}(3, 3, 1, 1, 1)$ with entries 1, 2 in row 1 (i.e. T SR-at-1), $p_T(q) = 0$ if and only if at least one of the following holds:*

- (α) *entries 3, 4 both in column 1, equivalently $T(3, 1) = 4$;*
- (β) *entries 3, 5, 6 in column 1, equivalently $T(2, 1) = 3, T(3, 1) = 5, T(4, 1) = 6$;*
- (γ) *entries 7, 8 in column 1 but 9 not in column 1 above the bottom, equivalently $T(4, 1) = 7, T(5, 1) = 8$ with the constraint forcing $9 \in (2, 3)$ (so $T(5, 1) = 8$ is below $T(4, 1) = 7$, and 9 sits in row 2 not in column 1).*

This statement was verified on all 50 SR-at-1 tableaux with 0 counterexamples.

4.2 Arithmetic structure

The discriminating SC-pair indices are

$$\{3, 5, 7\} = \{q, q + 2, q + 4\}, \quad q = 3.$$

This is the column-1 promotion arithmetic of the SRD chain $\Omega = R'_3 \cdots R'_9$, with column-1 transitions spaced by 2. The three clauses (α), (β), (γ) correspond to which factor of Ω , in the right-to-left order $R'_9, R'_8, \dots, R'_3$, first annihilates the running vector $\Omega_{\geq k} v_T$:

clause	col-1 SC pair	first-kill R'_k	count
α	(3, 4) at indices (2, 1), (3, 1)	R'_8	20
β	(5, 6) at indices (3, 1), (4, 1)	R'_7	9
γ	(7, 8) at indices (4, 1), (5, 1), $9 \notin \text{col } 1$	R'_6	6

The three counts $20 + 9 + 6 = 35$ match the residual cardinality exactly.

4.3 Refuted simpler conjectures

Several simpler discriminators were tested against the same data and refuted.

(Conj A) SC index threshold. Try $p_T = 0 \Leftrightarrow \min(\text{SC}(T)) \leq t$ or $\max(\text{SC}(T)) \leq t$ for some uniform t . Refuted: on the 50 SR-at-1 tableaux, the zero-set values of $\min \text{SC}$ are $\{3, 4, 5, 6, 7\}$ and the nonzero-set values are $\{4, 5, 6, 7, 8\}$; the overlap kills any monotone-threshold rule in either direction. The same overlap occurs with $\max \text{SC}$.

(Conj B) SC pair with prefix $\{1, \dots, i\}$ filling row 1. Try $p_T = 0 \Leftrightarrow T$ has an SC pair at some index i such that $\{1, 2, \dots, i\}$ fills row 1 from the left. Vacuous on SR-at-1: row 1 has length 3, so the prefix condition demands $i \leq 3$; SR-at-1 forbids SC at $i = 1$; SC at $i = 2$ requires 2 directly above 3 in column 2, impossible when row 1 starts $1, 2, X$ with $X \geq 3$. The conjecture predicts $p_T = 0$ on no SYT and is therefore false on all 35 residual tableaux.

(Conj C) Row-1 small-entry condition. Try $p_T = 0 \Leftrightarrow$ either row 1 of T equals 1, 2, 3, or row 1 starts 1, 2 and row 2 starts 3, 4 with cells (2, 1), (2, 2). Refuted from both sides: among the 35 residual tableaux, 25 satisfy *neither* sub-clause of (Conj C), and among the 10 tableaux satisfying one of the sub-clauses, 5 are nonresidual. This conjecture, originally suggested by the [[omega-survivor-pattern]] memory note, captures only a small subset of the residual.

4.4 Verification details

The discriminator was checked tableau-by-tableau, recording for each T the column-1 entries $T(2, 1), T(3, 1), T(4, 1), T(5, 1)$, the SC list, the descent set, and the first-vanish stage of Ωv_T under the right-to-left factorisation $\Omega = R'_3 \cdots R'_9$. The 35 residual tableaux partition cleanly into the three clauses with no overlap (no tableau satisfies two of (α, β, γ) simultaneously in the verification data), and the 15 nonresidual SR-at-1 tableaux satisfy none of (α, β, γ) except for the three sign-cancellation cases discussed in Section 5.

The verification script and full table are at `2026-05-19-V33111-residual-diagnostic.py` and `2026-05-19-V33111-residual-diagnostic.md`.

5 The sign-cancellation phenomenon

The three SYTs T_0, T_{10}, T_{20} have entries 7, 8, 9 all stacked in column 1 (rows 3, 4, 5). They therefore satisfy *both* the SC pair (7, 8) in column 1 and the SC pair (8, 9) in column 1. The naive γ -clause without the exclusion “9 not in column 1” would predict $p_T = 0$ on these three; in fact $p_T \neq 0$ for all three.

- $T_0 = (1, 2, 3 \mid 4, 5, 6 \mid 7 \mid 8 \mid 9)$: SC = [7, 8], $p_{T_0} \neq 0$.
- $T_{10} = (1, 2, 4 \mid 3, 5, 6 \mid 7 \mid 8 \mid 9)$: SC = [7, 8], $p_{T_{10}} \neq 0$.
- $T_{20} = (1, 2, 5 \mid 3, 4, 6 \mid 7 \mid 8 \mid 9)$: SC = [5, 7, 8], $p_{T_{20}} \neq 0$.

Interpretation: the column-1 SC pair (7, 8) contributes a -1 -eigenvalue absorption in R'_8 , which on its own would force $R'_6 \cdots R'_8 v_T \in \ker S_7$. But the further SC pair (8, 9) in column 1 contributes an additional -1 -absorption in R'_9 , and the two interfere so that the net partial product $R'_6 \cdots R'_9 v_T$ no longer lies in $\ker S_7$. The “non-continuation” clause in (γ) is therefore essential: it forbids the SC pair (8, 9) stacked immediately below (7, 8) in column 1, breaking the cancellation.

The analogous neutralisation does not occur for α or β because the candidate “next” SC pairs (4, 5) in column 1 (for α) and (6, 7) in column 1 (for β) are never realised in any SR-at-1 tableau of this shape: the SR-at-1 condition together with the column-1 length-5 constraint forces $T(3, 1) \in \{4, 5, 6, 7, 8\}$ varying, and $T(4, 1) > T(3, 1)$ with $T(5, 1) = 9$. In particular $T(3, 1) = 4 \Rightarrow T(4, 1) \in \{5, 6, 7, 8\}$, so the pair (4, 5) in column 1 is realised only when $T(4, 1) = 5$ alongside $T(3, 1) = 4$, but then the pair (5, 6) in column 1 would require $T(5, 1) = 6 \neq 9$, contradiction. A similar contradiction rules out (6, 7) stacked immediately below (5, 6) in column 1. The asymmetry between (γ) and (α, β) is therefore an artefact of 9 being the largest entry and naturally landing at the bottom of column 1.

6 Open questions

6.1 Generalisation to other $\hat{\lambda}$

Theorem 5 is verified on one shape. The arithmetic structure “column-1 SC pairs at indices in $\{q, q+2, q+4, \dots\}$ ” suggests a uniform rule:

Conjecture 6 (Column-1 SC discriminator, general form). *For $\lambda = (\hat{\lambda}, 1^q)$ with $\hat{\lambda}$ admitting an SR-at-1 stratum and $\tau(\hat{\lambda}) = 0$, an SR-at-1 tableau T satisfies $p_T = 0$ if and only if T has an SC pair in column 1 at some index $i \in \{q, q+2, \dots\}$ within the column-1 length range, subject to a non-continuation condition at the bottom of column 1.*

The smallest accessible test is $\lambda = (3, 3, 2)$, where the prior probe found 5 residual tableaux all satisfying SC at index 3 (the column-1 start for $\hat{\lambda} = (3, 3, 1)$); this is consistent with Conjecture 6 but vacuous in the discriminator’s nontrivial direction. The next informative test is $\lambda = (4, 3, 1, 1, 1)$ or $\lambda = (3, 3, 1, 1, 1, 1)$, where the column-1 promotion arithmetic predicts spacing 2 but at shifted base.

6.2 Structural proof

The discriminator should follow from a Hecke-algebraic identity of the form: *the partial chain product $R'_{i+1}R'_{i+2} \cdots R'_n v_T$ lies in $\ker S_i$ whenever T has an SC pair in column 1 at index $i \in \{q, q+2, q+4\}$ without continuation.* For (α) with $i = 3$, this asks that $R'_4 \cdots R'_9 v_T \in \ker S_3 = \text{span}\{v_S : S \text{ has SC at } 3\}$ when $T(2, 1) = 3, T(3, 1) = 4$. The natural route is to analyse how R'_k acts on the position of the letter $i+1$ in the support of the running vector (a “letter-shift” argument, by analogy with the SC-at-1 trivial proof where R'_n has S_1 at the right end).

A structural proof would also explain the asymmetric clause in (γ) : the non-continuation $9 \notin$ column 1 should arise as a transversality condition on the letter-shift dynamics at the column-1 boundary.

6.3 Correction to the memory note

The memory note [[omega-survivor-pattern]] flagged a count of 12 Ω^* -only zeros on $(3, 3, 1, 1, 1)$, characterised by row 1 = 1, 2, 3 (11 cases) or row 1 starting 1, 2 with row 2 starting 3, 4 adjacent (1 case). The direct probe finds 0 Ω^* -only cases on this shape: all 35 residual zero diagonals satisfy $\Omega v_T = 0$ and $\Omega^* v_T = 0$ simultaneously. The 12 count is retracted for $\lambda = (3, 3, 1, 1, 1)$. The original observation likely referred to a different shape or to a finer right-kernel-only stratum within the residual; this requires cross-checking against the shape the memory was originally derived from.

The corrected accounting on $\lambda = (3, 3, 1, 1, 1)$ is:

SR-at-1 residual ($p_T = 0$)	35
of which $\Omega v_T = 0$	35
of which $\Omega^* v_T = 0$	35
of which Ω^* -only (i.e. $\Omega v_T \neq 0$)	0

6.4 Why $\{3, 5, 7\}$?

The set $\{3, 5, 7\}$ equals $\{q + 2k : k = 0, 1, 2\}$ with $q = 3$. The chain factors $R'_3, R'_4, \dots, R'_9$ index a sequence of column-1 transition steps in the SRD reduction. The killing stages $\{R'_6, R'_7, R'_8\}$ are the *middle* of the chain, not the endpoints; R'_3 (leftmost) and R'_9 (rightmost) never appear as first-kill stages on the residual. This middle-of-chain symmetry is suggestive but currently not exploited.

Acknowledgements

This note builds on the SC-at-1 trivial reduction (2026-05-19-sc-at-1-trivial-reduction.tex), the SR-conjecture refutation (2026-05-19-sr-conjecture-refuted.tex), and the two-sided kernel dichotomy (2026-05-18-d-class-row-zero.tex). The Hoefsmit basis machinery is at 2026-05-08-rank-Pi/v diagonal probes at 2026-05-19-V33111-residual-diagnostic.py; cell-partition data at 2026-05-19-V332-V3311

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